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Example at the end of last class

e.g. $f(x, y, z) = x^2 + y^2 + z$ tangent hyperplane to f at $(1, 1, 2)$

↳ $w = f(x, y, z)$ gives a hypersurface inside $\mathbb{R}^4$

$w = f(1, 1, 2) + \vec{\nabla} f|_{(1, 1, 2)} \cdot (\vec{x} - \vec{a})$ gives a hyperplane tangent to $w = f(x, y, z)$ at $\vec{a}$.

$$\vec{\nabla} f(x, y, z) = (2x, 2y, 1)$$

$$\vec{\nabla} f|_{(1, 1, 2)} = (2, 2, 1)$$

$$w = 4 + (2, 2, 1) \cdot (x-1, y-1, z-2)$$

$$w = f(\vec{a}) + \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a})$$

The differential or Jacobian matrix

How do we treat differentiability of a function with several components?

$$F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n \quad \text{we can write as } F(\vec{x}) = \begin{pmatrix} F_1(\vec{x}) \\ F_2(\vec{x}) \\ \vdots \\ F_n(\vec{x}) \end{pmatrix}$$

def: We say $F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ is differentiable at an interior point $\vec{a}$ in X if there is a linear function $H: \mathbb{R}^m \rightarrow \mathbb{R}^n$ of the form

$$H(\vec{x}) = F(\vec{a}) + A(\vec{x} - \vec{a}) \quad (\text{where } A \text{ is a matrix in } M_{n \times m})$$

$$\text{such that } \lim_{\vec{x} \rightarrow \vec{a}} \frac{F(\vec{x}) - H(\vec{x})}{\|\vec{x} - \vec{a}\|} = 0.$$

Theorem: $F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ is differentiable at $\vec{a}$ if and only if each component function F_j is differentiable at $\vec{a}$ (for $j=1, \dots, n$).

def: If $F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$, we call the matrix of partial derivatives

$$DF(\vec{a}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1}(\vec{a}) & \dots & \frac{\partial F_1}{\partial x_m}(\vec{a}) \\ \frac{\partial F_2}{\partial x_1}(\vec{a}) & \dots & \frac{\partial F_2}{\partial x_m}(\vec{a}) \\ \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial x_1}(\vec{a}) & \dots & \frac{\partial F_n}{\partial x_m}(\vec{a}) \end{bmatrix}$$

the differential or

the Jacobian matrix.

* DF has: as many rows as F components
as many columns as independent variables

e.g.: $F(x, y) = (x y, x^2, e^{xy})$

$F_1 \uparrow \quad F_2 \uparrow \quad F_3 \uparrow$

dimension of DF: 3×2

how many rows: 3
how many columns: 2

$$\begin{bmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \\ \frac{\partial F_3}{\partial x} & \frac{\partial F_3}{\partial y} \end{bmatrix} = \begin{bmatrix} y & x \\ 2x & 0 \\ ye^{xy} & xe^{xy} \end{bmatrix} = DF(x, y) = DF|_{(x, y)}$$

e.g.: $F(x, y, z) = (2x + y, z - 3x, y + 2x, x)$

dimension of DF: 4×3

$F: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ linear transform.

$$DF|_{(x, y, z)} = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$F\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Practical tool

Theorem: If all partial derivatives of all components of $F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ exist and are continuous in a neighbourhood of $\vec{a}$ in X , then F is differentiable at $\vec{a}$ and its best linear approximation around $\vec{a}$ is

$$H(\vec{x}) = F(\vec{a}) + DF|_{\vec{a}} (\vec{x} - \vec{a}) .$$

e.g. $F(x, y, z) = (\cos(xy), z e^{xy})$

$H(\vec{x})$ linear approx
at $x=\pi \quad y=1 \quad z=2$

$$DF(x, y, z) = \begin{bmatrix} -y \sin(xy) & -x \sin(xy) & 0 \\ y z e^{xy} & x z e^{xy} & e^{xy} \end{bmatrix}$$

$$\vec{a} = (\pi, 1, 2)$$

$$H(\vec{x}) = F(\vec{a}) + DF|_{\vec{a}} (\vec{x} - \vec{a})$$

$$= \begin{pmatrix} -1 \\ 2e^\pi \end{pmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ ze^\pi & 2\pi e^\pi & e^\pi \end{bmatrix} \begin{pmatrix} x - \pi \\ y - 1 \\ z - 2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 \\ ze^\pi + 2e^\pi(x - \pi) + 2\pi e^\pi(y - 1) + e^\pi(z - 2) \end{pmatrix}$$

Differentiation Laws

Theorem: Suppose

- 1) $F: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ diff. at $\vec{a}$
- 2) $G: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ diff. at $\vec{a}$
- 3) $f: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ diff. at $\vec{a}$
- 4) $g: X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ diff. at $\vec{a}$

Then,

- 1) $F + G$ is diff. at $\vec{a}$ and $D(F+G)|_{\vec{a}} = DF|_{\vec{a}} + DG|_{\vec{a}}$
- 2) if $k \in \mathbb{R}$, then kF is diff. at $\vec{a}$ and $D(kF)|_{\vec{a}} = k DF|_{\vec{a}}$
- 3) $f \cdot g$ is differentiable at $\vec{a}$ and $D(fg)|_{\vec{a}} = g(\vec{a})Df|_{\vec{a}} + f(\vec{a})Dg|_{\vec{a}}$
- 4) if $g(\vec{x}) \neq 0$, then $\frac{f}{g}$ is diff. at $\vec{a}$ and $D\left(\frac{f}{g}\right)|_{\vec{a}} = \frac{g(\vec{a})Df|_{\vec{a}} - f(\vec{a})Dg|_{\vec{a}}}{(g(\vec{a}))^2}$

Some functions that are differentiable on all their domain

Polynomials, exponential, logarithm $\ln$

trig functions

Higher order derivatives

We can take partial derivatives of partial derivatives and then we call 2^{nd} order partial derivatives

e.g. $f(x, y) = x^2y + e^{x+y}$

compute 2^{nd} order partial derivatives

$$\frac{\partial f}{\partial x}(x, y) = 2xy + e^{x+y}$$

$$\frac{\partial f}{\partial y}(x, y) = x^2 + e^{x+y}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) (x, y) = 2y + e^{x+y}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) (x, y) = e^{x+y}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) (x, y) = 2x + e^{x+y}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) (x, y) = 2x + e^{x+y}$$

these are called crossed or mixed
 2^{nd} order partial derivatives

Thm: if $f \in C^2$ (all 2^{nd} order part. der. are continuous)
then $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$.

e.g. Verify that $f(x, y, z) = 2x^2 - y^2 - z^2$
satisfies $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$.